



Typology problems in Data Science:

- (a) Supervised Learning Problem: predict based on input features
 - Classification: Categorical data (eg Spam Detection, Image Recognition)
 - Regression: Continuous outcome (eg house Prediction, sales forecasting)
- (b) Unsupervised Learning Problems: data without labelled outputs
 - Clustering: Grouping similar data points (eg. Customer Segmentation)
 - Dimensionality Reduction - Reducing no. of features while preserving important info (eg PCA, t-SNE)
- (c) Reinforcement: Reward desired behaviour (eg. game playing)
- (d) Descriptive & Diagnostic Analytics
 - Descriptive: Summarizing historical data (eg Sales Report)
 - Diagnostic: Understanding why something happened
- (e) Prescriptive & Predictive:
 - Predictive: forecasting future (Demand Prediction)
 - Prescriptive: Recommending actions (Optimization problem)

LINEAR ALGEBRA

Vector & Matrices

Matrix Operations

Application

→ PCA

→ Neural Network & Deep learning

→ Solving system of equation in optimization

STRUCTURE OF SOLVING DATA SCIENCE PROBLEM with Ex
 Problem: Predict whether a customer will leave a
 Subscription service. (Churn)

1. Define the Problem: Predict churn (Binary classification)
2. Understand the data Analyse Customer Data
3. Formulate hypothesis Customer with low usage are more likely to churn
4. Choose Tools Use logistic regression or decision tree
5. Build & Evaluate Train the model & evaluate using accuracy & ROC-AUC
6. Iterate Try different features or algo to improve
7. Communicate Show insight with business team

Linear Algebra: Matrix

- Singular $\det(A) = 0$ (does not have inverse, No solⁿ or infinite solⁿ)
- Non Singular $(\det(A) \neq 0)$ (has inverse, has unique solⁿ)

Matrices & Their Properties

1) determinant:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Determinant:

Scalar value that provides info about matrix invertibility & volume scaling factor.

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

If $\det(A) = 0$ A is Singular (not invertible)
 not invertible means can't be inverted

2) Trace - is the sum of its diagonal elements.
 for A, $\text{tr}(A) = a_{11} + a_{22} + a_{33}$

3) Rank - is the maximum number of linearly independent rows or column.
 for Eg:

Rank of A, is number of non zero rows in its row echelon form.

Ex:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 4R_1$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix}$$

Subtract 7 times first row from 3rd Row

$$R_3 \rightarrow R_3 - 7R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank A is 2

By two Methods

(1) Gaussian Elimination (Row Echelon form)

(2) Determinant Method

If determinant is non zero, the rank is full

If determinant is zero rank less than 3

Application of Rank in Data Science

1. Dimensionality Reduction
2. Collinearity Detection
3. Recommendation System
4. NLP
5. Computer Vision
6. Network Analysis

RANK NULLITY THEOREM

$\text{RANK}(A) + \text{Nullity}(A) = \text{No. of Columns of } A$ also known as Kernel

Nullity: of a matrix is dimension of its null space
(Solⁿ to $Ax=0$)

for A , nullity is given by,

$$\text{nullity}(A) = n - \text{rank}(A)$$

↳ no. of columns

Ex:

$$\text{nullity}(A) = 3 - 2 = 1$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

⇒ Infinite Solⁿ, meaning null space is nontrivial

Applications:— (Used in feature selection to find Redundant features)
Collinearity detection, Underdetermined System, Regularization,
Network Analysis, Signal Processing

Eigen Values & Eigen Vectors

To understand this we need to know:

(i) Transpose of a Matrix

$$(A^T)_{ji} = A_{ij}$$

Ex $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

$$A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

- Used in Covariance Matrices to find relationship b/w variables
- Used in PCA to transform data into new coordinate system

(ii) Inverse of a Matrix

$$AA^{-1} = A^{-1}A = I \rightarrow \text{Identity Matrix}$$

A matrix is invertible if $\det(A) \neq 0$

Ex:

$$A = \begin{bmatrix} 2 & 4 & 1 \\ 3 & 1 & 5 \\ 7 & 2 & 6 \end{bmatrix}$$

$$\det(A) = 2(1 \times 6 - 5 \times 2) - 4(-11) + 1(4) = 35 \neq 0$$

- Used in Linear Regression to solve for coefficients using Normal equations.
- Used in Graph theory to compute shortest path.

Inverse Ex:-

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$$

adjugate (Transpose of the Cofactor matrix)

$$A = \begin{bmatrix} 2 & 4 & 1 \\ 3 & 1 & 5 \\ 7 & 2 & 6 \end{bmatrix}$$

$$|A| = 59 \neq 0 \text{ hence invertible}$$

Find Cofactor Matrix

$$C = \begin{bmatrix} \begin{vmatrix} 1 & 5 \\ 2 & 6 \end{vmatrix} & -\begin{vmatrix} 3 & 5 \\ 7 & 6 \end{vmatrix} & \begin{vmatrix} 3 & 1 \\ 7 & 2 \end{vmatrix} \\ -\begin{vmatrix} 4 & 1 \\ 2 & 6 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 7 & 6 \end{vmatrix} & -\begin{vmatrix} 3 & 1 \\ 7 & 2 \end{vmatrix} \\ \begin{vmatrix} 4 & 1 \\ 1 & 5 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 3 & 5 \end{vmatrix} & \begin{vmatrix} 2 & 4 \\ 3 & 1 \end{vmatrix} \end{bmatrix}$$

$$C = \begin{bmatrix} -4 & +17 & -1 \\ -22 & 5 & +24 \\ 19 & -7 & -10 \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} -4 & -22 & 19 \\ 17 & 5 & -7 \\ 1 & 24 & -10 \end{bmatrix}$$

$$A^{-1} = \frac{1}{59} \begin{bmatrix} -4 & -22 & 19 \\ 17 & 5 & -7 \\ 1 & 24 & -10 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -0.0678 & -0.3729 & 0.3220 \\ 0.2881 & 0.0847 & -0.1186 \\ -0.0169 & 0.4068 & -0.1695 \end{bmatrix}$$

Used in PCA for dimensionality Reduction
in Spectral clustering for Grouping data.



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Eigen Values & Eigen Vector

for Matrix A , an eigen vector v & eigen value λ satisfy:

$$Av = \lambda v$$

→ helps in understanding Linear transformations.

A is $n \times n$ matrix

v is nonzero vector

λ is a scalar (eigen value)

it means that matrix A transforms v , it only scales it by λ without changing its direction.

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Step 1. Compute characteristic eqⁿ:

$$\det(A - \lambda I) = 0$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (2-\lambda) \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} - (-1) \begin{vmatrix} -1 & -1 \\ 0 & 2-\lambda \end{vmatrix} + 0 \begin{vmatrix} -1 & 2-\lambda \\ 0 & -1 \end{vmatrix}$$

$$2\lambda^2 - 8\lambda + 6 - \lambda^3 + 4\lambda^2 - 3\lambda - 2 + \lambda = 0$$

$$-\lambda^3 + 6\lambda^2 - 10\lambda + 4 = 0$$

$$\lambda_1 = 3.414 \quad \lambda_2 = 2 \quad \lambda_3 = 0.586$$

Step 2: Compute Eigen Vectors

Solve:

$$(A - \lambda I)v = 0$$

For $\lambda_2 = 2$:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We get $v_2 = (1, 0, -1)$

Simultaneously we find eigen vectors for λ_1 & λ_3

MATRIX FACTORIZATION



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Some Concepts :

Dot Product of two vectors : - (inner product)

$$A \cdot B = A_1 B_1 + A_2 B_2 + \dots + A_n B_n$$

$$A \cdot B = |A| |B| \cos \theta$$

$$A = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad B = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad A \cdot B = (3 \times 2) + (4 \times -1) \\ = 6 - 4 = 2$$

Magnitude (~~the~~ Norm) of a Vector : -

Euclidean norm of a vector $A = (A_1, A_2, \dots, A_n)$

$$\text{is } |A| = \sqrt{A_1^2 + A_2^2 + \dots + A_n^2}$$

$$|A| = \sqrt{3^2 + 4^2} = 5$$

angle b/w two vectors

$$\cos \theta = \frac{A \cdot B}{|A| |B|} \Rightarrow \theta = \cos^{-1} \left(\frac{A \cdot B}{|A| |B|} \right)$$

Linear Combination of Vectors : -

$$C_1 v_1 + C_2 v_2 + \dots + C_n v_n$$

$C_1, C_2, \dots, C_n$ are scalars (Coefficients)

$v_1, v_2, \dots, v_n$ are vectors

Ex :

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Linear Comb.

$$2v_1 + 3v_2 = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 9 \\ 12 \end{bmatrix} = \begin{bmatrix} 11 \\ 16 \end{bmatrix}$$

Linearly Independent :

Vectors $v_1, v_2, \dots, v_n$ are linearly independent if no vector in set can be written as linear combination of others.

Linearly Dependent - If at least one vector can be written as linear combination of others.

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$v_2 = 2v_1$$

Hence linearly dependent

Orthogonal Vectors

Vector A & B are orthogonal if their dot product is zero

$$A \cdot B = 0$$

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad B = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$A \cdot B = (1 \times -2) + (2 \times 1) = -2 + 2 = 0$$

Hence vectors are orthogonal.

Ortho normal Vector : if they are

1. Orthogonal to each other ($A \cdot B = 0$)
2. Unit Vector (each has magnitude 1)

$$\text{Ex } A = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad B = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$A \cdot B = 0$$

$$|A| = \sqrt{1/2 + 1/2} = 1 \quad |B| = 1$$

Projection : of a vector u onto v is :

$$\text{proj}_v u = \left(\frac{u \cdot v}{v \cdot v} \right) v$$

$$u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$v = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\text{proj}_v u = \left(\frac{32}{77} \right) \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Gram-Schmidt Orthogonalization

is a process / algorithm that converts a set of linearly independent vectors into an orthogonal set. This is useful in applications such as QR decomposition, PCA, & dimensionality Reduction.

Given a set of linearly independent vectors $v_1, v_2, \dots, v_n$ the goal is to generate an orthonormal basis $q_1, q_2, \dots, q_n$.

1. Start with q_1 , the normalized first vector:

$$q_1 = \frac{v_1}{\|v_1\|}$$

2. Compute next orthogonal vector by removing the component of v_2 along q_1 :

$$u_2 = v_2 - \text{proj}_{q_1}(v_2)$$

$$\text{where } \text{proj}_{q_1}(v_2) = \frac{q_1 v_2}{q_1 q_1} (q_1)$$

Normalize u_2 to get q_2

$$q_2 = \frac{u_2}{\|u_2\|}$$

3. Repeat for all vectors:

$$u_k = v_k - \sum_{j=1}^{k-1} \text{proj}_{q_j}(v_k)$$

Normalize:

$$q_k = \frac{u_k}{\|u_k\|}$$



Ex: -

$$v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Step 1. Compute q_1

$$q_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{3^2+1^2}} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Step 2 Compute q_2

$$\begin{aligned} \text{find } \text{proj}_{q_1}(v_2) &= \left(\frac{q_1 \cdot v_2}{q_1 \cdot q_1} \right) q_1 \\ &= \left(\left(\frac{3}{\sqrt{10}} \cdot 2 \right) + \left(\frac{1}{\sqrt{10}} \cdot 2 \right) \right) \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix} \\ &= \frac{8}{\sqrt{10}} \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix} \end{aligned}$$

$$u_2 = v_2 - \text{proj}_{q_1}(v_2)$$

$$= \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 2.4 \\ 0.8 \end{bmatrix} = \begin{bmatrix} -0.4 \\ 1.2 \end{bmatrix}$$

Normalise u_2

$$\|u_2\| = \sqrt{0.16 + 1.44} = \sqrt{1.6} = 1.26$$

$$q_2 = \frac{u_2}{\|u_2\|} = \begin{bmatrix} -0.32 \\ 0.95 \end{bmatrix}$$

FINAL
ORTHONORMAL BASIS

$$q_1 = \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}$$

$$q_2 = \begin{bmatrix} -0.32 \\ 0.95 \end{bmatrix}$$

NORM is a function that measures the "size" or "length" of a vector in a vector space. It is used in various mathematical & ML applications to quantify distance, similarity & stability.

A norm $\|x\|$ satisfies the following properties for any vector x & scalar α :

1. Non-negativity.

$$\|x\| \geq 0 \text{ \& } \|x\| = 0 \text{ if \& only if } x = 0$$

2. Scalar Multiplication

$$\|\alpha x\| = |\alpha| \|x\|$$

3. Triangle Inequality

$$\|x + y\| \leq \|x\| + \|y\|$$

4. Subadditivity

Norm should satisfy the properties of a metric

Types of Norms

1. L_1 Norm (Manhattan Norm or Taxicab Norm)

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

Measures the sum of absolute values of the vector components.

Used in Lasso Regression (L_1 Regularisation) & Sparse Models.

$$x = [3, -4, 5]$$

$$\|x\|_1 = |3| + |-4| + |5|$$

$$= 3 + 4 + 5$$

$$= 12$$

Help in feature selection by forcing some coefficient to become zero

2. L_2 Norm (Euclidean Norm / Magnitude of Vector)

It calculates the square root of the sum of squared components.

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

Ex: $x = [3, -4, 5]$

$$\|x\|_2 = \sqrt{3^2 + (-4)^2 + (5)^2} = \sqrt{50} \approx 7.07$$

Applications :-

- Ridge Regression (L_2 Regularisation) by penalizing large weights
- KNN : find closest neighbor
- PCA - Finds direction of maximum variance

3. L_∞ Norm (Maximum Norm / Chebyshev Norm)

It takes largest absolute value among vector components.

$$\|x\|_\infty = \max |x_i|$$

Ex.

$$x = [3, -4, 5]$$

$$\|x\|_\infty = \max(|3|, |-4|, |5|) = 5$$

Application :

Used in Robust Optimization to find worst case scenario

Outlier detection - helps in anomaly detection. (as outlier usually have large values)

4. L_p Norm (Generalized p-norm)

A general formula for compound norms:

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

Ex: ($p=3$) $x = [3, -4, 5]$

$$\begin{aligned} \|x\|_3 &= (|3|^3 + |-4|^3 + |5|^3)^{1/3} \\ &= (27 + 64 + 125)^{1/3} \\ &\approx 5.34 \end{aligned}$$

Application :

- Tunable norms
- Used in deep learning loss fn & optimization

5. Frobenius Norm (for Matrices)

This is Euclidean Norm for matrices :-

$$\|A\|_F = \sqrt{\sum_{i,j} A_{i,j}^2}$$

Ex :

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\|A\|_F = \sqrt{1^2 + 2^2 + 3^2 + 4^2} = \sqrt{1+4+9+16} = \sqrt{30}$$

≈ 5.48

Used in Low Rank Approx. & PCA
Image compression Using Matrix factorization

6. Nuclear Norm (Sum of Singular Values).

This is used for low-rank Approximations!

$$\|A\|_* = \sum \sigma_i$$

→ singular values of matrix

→ Used Matrix Completion Problem : Used in Netflix Recommendation system

→ Compressed Sensing : efficient Signal Processing.

Hyperplane & half Plane

A hyperplane in $\mathbb{R}^n$ is an $(n-1)$ dimensional subspace:

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

half plane is one of the two region divided by hyperplane
Ex:-

In $\mathbb{R}^3$: $2x + 3y - z = 4$ defines hyperplane

Inequality $2x + 3y - z \leq 4$ defines half plane

MATRIX FACTORIZATION

it is a fundamental technique in linear Algebra that decomposes a matrix into a product of two or more matrices

Widely used in data science for dimensionality reduction, noise reduction & extracting latent features from data.

decompose a matrix A (of size $m \times n$) into two or more matrices such that:

$$A \approx B \cdot C$$

$m \times n \quad m \times k \quad k \times n$

Goal is to approximate A as closely as possible using lower-dimensional matrices B & C .

Types of Matrix Factorization:

1. Singular Value Decomposition (SVD)

(i) Full SVD (ii) Reduced SVD

2. Non-negative Matrix Factorization

3. QR decomposition

4. LU Decomposition

5. Eigen Decomposition